MIL-S-63164(AR)
25 June 1982
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

SUPPORT, GRAIN
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for grain supports used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-14	Molding, Plastic and Molded Plastic Parts, Thermosetting
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (*DD Form 1426*), appearing at the end of this document, or by letter.

FSC 1110

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STANDARDS

MILITARY

MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

11829174	Support, Grain

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The grain support shall comply with all requirements specified on drawing (dwg.) 11829174, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. The contractor shall provide objective evidence (see 6.4) that materials comply with applicable dwgs. and specifications.

3.3 First article. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

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3.4 Workmanship. Grain supports shall be fabricated in a thoroughly workmanlike manner and there shall be no evidence of cracks, chipped edges, inclusions, blisters or any other obvious plastic defects. Finished parts shall be clean and free of dirt, grease, dust and any other foreign matter. Cleaning methods shall not be injurious to the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.3), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Support, Grain	11829174	6

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspections specified in the first article

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inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source of supply as will be used in regular production.

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FIRST ARTICLE INSPECTION

MIL-S-63164(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Support, Grain	1 OF 1	11829174
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of Defects and Tests	6	---	3.1, 3.2 3.4	4.4.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot of parts shall not exceed one days production and shall be produced from a single lot of approved material (see 3.2).

4.4.2 Examinations. See MIL-A-48078 and the Quality Conformance Inspection Tables herein.

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QUALITY CONFORMANCE INSPECTION

MIL-S-63164(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Support, Grain	1 OF 1	11829174
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Outside diameter	4.4.3.1		3.1	Gage/TMC
102.	Inside diameter of hole	4.4 3.1		3.1	Gage/TMC
103.	Location inside diameter of hole	4.4 3.1		3.1	Gage/TMC
104.	Radius improper	4.4 3.1		3.1	Gage/TMC
105.	Overall thickness	4.4 3.1		3.1	Gage/TMC
106.	Flatness of face	4.4 3.1		3.1	Gage/TMC
107.	.04 flat	4.4 3.1		3.1	Gage/TMC
108.	Surface finish missing or improper		100%	3.1	Visual
109.	Part number improper or missing		100%	3.1	Visual
110.	Material per MIL-M-14 (Type GPI-100)		---	3.2	Visual (see 6.4)
111.	Poor workmanship		100%	3.4	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Packaging	1 OF 1	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		-	5.0	Visual
102.	Items not clean and free of contamination and foreign material prior to packaging		100%	5.1.1	Visual
103.	Separator cushions missing		100%	5.1.1	Visual
104.	Spiral container improper or improperly sealed		100%	5.1.1	Visual
105.	Filler and cushioning does not provide firm pack		100%	5.1.1	Visual/Manual
106.	Exterior pack improper or not properly sealed		100%	5.1.2	Visual
107.	Marking on spiral wound or exterior containers missing, improper or illegible		100%	5.1.3	Visual
108.	Evidence of poor workmanship		100%	-----	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and herein.

4.4.3.1 Tool manufacturing control. When tool manufacturing control (TMC) is specified, the first five pieces initially produced from each mold cavity inspected (after a minimum mold preheat of one hour) for the applicable classification of defects herein shall constitute initial mold acceptance. Subsequently, the first and last five pieces from each mold cavity from each days production (after a minimum mold preheat of one hour) inspected for the applicable classification of defects herein shall constitute lot acceptance. In addition, any time the mold requires repair an additional five pieces from each mold cavity (after a minimum mold preheat of one hour) shall be inspected for the applicable classification of defects herein before resumption of production. If any sample fails to meet any of the applicable inspection requirements, the lot of production under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods. Not applicable.

5. PACKAGING

5.1 Packaging and packing. Packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be semi-bulk packaged in moistureproof spirally wound containers in accordance with MIL-STD-1169, Level A. Items shall be separated by cushions and adequate filler and cushioning shall be provided to assure a firm pack and prevent cracking, chipping or other physical damage during handling and transit.

5.1.2 Packing. Spiral wound containers shall be packed in accordance with MIL-STD-1169, Level B.

5.1.3 Marking. Spiral wound containers and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no.

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c. Used on M753 Projectile
d. Manufacturer
e. Quantity enclosed
f. Lot no.
g. Contract or purchase order no.
h. Date of acceptance

6. NOTES

6.1 Intended use. Grain supports covered by this specification are intended for use in motor body bonding assemblies of rocket motors for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data rquirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

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6.4 Objective evidence for material. A statement (certification) supported by test data that each lot of molding compound meets the requirements of the applicable dwgs. and specifications.

6.5 Supersession data. This specification includes the requirements of Purchase Description AR-PD-19 dated 9 April 1979.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

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Preparing activity

Army-AR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-63164 SUPPORT, GRAIN M753 8 INCH PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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B. RECOMMENDED WORDING CHANGE

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